

RESEARCH ARTICLE

Who Upgrades the Plant? A Tripartite Evolutionary Game Model for PPCPs Removal Technology Adoption

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ABSTRACT

Pharmaceuticals and personal care products (PPCPs) contamination in aquatic environments has attracted considerable research attention, yet most studies focus on treatment technologies while the governance mechanisms coordinating stakeholder behavior remain underexplored. This study develops a tripartite evolutionary game model to analyze strategic interactions among pharmaceutical enterprises, wastewater treatment plants, and government environmental agencies in PPCPs pollution control. Replicator dynamics and stability analysis identify six stable equilibrium configurations and reveal that complete cooperation equilibrium cannot achieve asymptotic stability due to government fiscal incentives for regulatory withdrawal. Results demonstrate that technology-upgrading decisions of treatment plants respond primarily to government subsidies and penalties rather than upstream enterprise behavior. The mathematical finding that upstream enterprise production choices do not directly alter treatment plant upgrading incentives through cost transmission channels reveals that market-based coordination cannot spontaneously emerge without regulatory intervention, thereby strengthening the theoretical case for sustained government engagement. Sensitivity analysis indicates that reducing clean production costs and enhancing green product revenues accelerate system convergence toward environmentally favorable equilibria. Additional sensitivity analysis of treatment plant subsidy and penalty parameters confirms that both positive incentives and negative sanctions effectively promote technology adoption when appropriately calibrated. These findings provide theoretical guidance for designing differentiated policy instruments that balance market incentives with sustained regulatory oversight.

1 | Introduction

Pharmaceuticals and personal care products (PPCPs) have become pervasive contaminants in aquatic ecosystems across the globe, raising significant concerns for both ecological integrity and human health (Kallenborn et al. 2018). These biologically active compounds are intentionally designed to elicit physiological responses at trace concentrations, a characteristic that renders them particularly problematic when they enter environmental compartments for which they were never intended. The primary pathway through which PPCPs reach surface waters is via wastewater treatment plant (WWTP) effluents, where conventional activated sludge processes often fail to achieve

adequate removal due to the recalcitrant molecular structures and low biodegradability of many pharmaceutical compounds (Tarpani and Azapagic 2018). This incomplete elimination transforms WWTPs from pollution control barriers into conduits for micropollutant discharge, creating a persistent source of contamination that affects receiving water bodies on a continuous basis.

The geographical scope of PPCP contamination spans from densely urbanized watersheds to remote polar regions. Monitoring campaigns conducted in the River Lambro basin, one of the most industrialized areas in Italy, detected pharmaceuticals and other emerging contaminants at concentrations

Summary

- Government subsidies drive treatment plant technology upgrading more than upstream pollution loads.
- Complete cooperation equilibrium remains structurally unstable without sustained regulatory oversight.
- Reducing clean production costs accelerates voluntary adoption of environmentally friendly strategies.
- Differentiated policy instruments outperform uniform penalties in PPCPs governance effectiveness.

reaching the microgram per liter range in untreated wastewater, with substantial mass loads persisting through treatment processes into final effluents (Castiglioni et al. 2018). Similar patterns emerge in mixed land use watersheds across North America, where contaminants including carbamazepine, sulfamethoxazole, and caffeine exhibit detection frequencies exceeding 70% in surface water samples, with concentrations influenced by proximity to WWTP discharge points and seasonal hydrological conditions (Fairbairn et al. 2016). Even the Arctic environment, long considered pristine, now harbors detectable residues of over 110 pharmaceutical substances in coastal seawater and biota, attributable largely to domestic wastewater discharges from settlements lacking modern treatment infrastructure (Kallenborn et al. 2018). This ubiquity underscores that PPCP pollution transcends local boundaries and demands coordinated responses across diverse governance contexts.

The ecological consequences of chronic PPCP exposure have attracted sustained investigative attention. Laboratory studies demonstrate that environmentally relevant concentrations of metformin, one of the most widely prescribed antidiabetic medications, induce endocrine disruption in adult male fathead minnows, manifesting as significant upregulation of vitellogenin messenger RNA, a biomarker typically associated with female reproductive physiology (Niemuth et al. 2015). Mesocosm experiments exposing three-spined stickleback populations to mixtures of commonly detected pharmaceuticals reveal impaired survival among female fish at higher treatment levels, alongside measurable impacts on condition factors at both individual and population scales (David et al. 2020). Marine organisms face analogous threats; tiered toxicity assessments of diclofenac exposure in the brown mussel *Perna perna* document decreased lysosomal membrane stability, DNA damage, and oxidative stress responses at concentrations as low as nanograms per liter, well within the range observed in coastal waters receiving submarine outfall discharges (Fontes et al. 2018). These findings collectively indicate that PPCPs exert subtle yet consequential effects on non-target organisms, with implications for ecosystem structure and function that may propagate through food webs.

The performance of treatment technologies in removing PPCPs varies considerably across compounds, processes, and operational conditions. Seasonal monitoring at Himalayan foothill WWTPs reveals negative removal rates for certain pharmaceuticals, meaning effluent concentrations exceed influent levels, a phenomenon attributed to deconjugation of metabolites and desorption from particulate matter during biological treatment (Silori et al. 2025). Comparative assessments of wastewater

treatment works demonstrate wide-ranging removal efficiencies for trace contaminants, with performance influenced by site-specific operational factors that remain incompletely characterized (Gardner et al. 2013). Advanced treatment technologies offer enhanced removal capabilities; membrane bioreactors achieve selective elimination of antibiotics through combined biodegradation and adsorption mechanisms (Zheng et al. 2019), while advanced oxidation processes (AOPs) and granular activated carbon filtration attain removal efficiencies exceeding 80% for many recalcitrant compounds (Risch et al. 2022). Yet life cycle assessments reveal that the direct water quality benefits obtained from advanced treatment are frequently outweighed by indirect environmental impacts stemming from elevated energy and resource demands, complicating straightforward recommendations for technology adoption (Risch et al. 2022). Multi-criteria decision-making frameworks have accordingly been developed to navigate these tradeoffs, incorporating factors such as hydraulic retention time and population equivalent alongside removal efficiency metrics to guide WWTP design decisions (Fernandez-Lopez et al. 2021).

Prioritization schemes represent another active research domain, seeking to identify which among thousands of pharmaceutical compounds warrant focused monitoring and regulatory attention. Approaches employing predicted environmental concentrations, persistence indicators, bioaccumulation potential, and ecotoxicological thresholds have been applied to rank pharmaceuticals according to their likelihood of causing environmental harm (Letsinger and Kay 2019). Risk-based prioritization of micropollutants detected in WWTP effluents across Chinese facilities identifies perfluorinated compounds, pesticide transformation products, and bisphenol A among the top-ranked substances, many of which are not currently subject to effluent discharge limits (Liu et al. 2024). The COVID-19 pandemic provided an unintended natural experiment illustrating how rapidly pharmaceutical consumption patterns translate into environmental contamination; wastewater surveillance documented dramatic surges in antipyretic concentrations, with ibuprofen levels increasing up to 67-fold and paracetamol up to 311-fold relative to pre-pandemic baselines, accompanied by elevated ecological risk quotients for aquatic organisms (Yu et al. 2025). Therapeutic agents specific to COVID-19 management, including remdesivir and dexamethasone, were detected persisting through treatment processes into effluents destined for beneficial reuse, introducing novel pharmaceutical compounds directly to agricultural soils and surface waters (Hayden et al. 2022). These observations highlight the dynamic coupling between human health interventions and environmental quality outcomes, a relationship mediated substantially by the decisions and capacities of wastewater treatment infrastructure.

Despite growing recognition of PPCPs as priority contaminants, existing governance frameworks exhibit structural deficiencies that constrain effective pollution control. Current regulatory architectures predominantly adopt end-of-pipe approaches that place remediation burdens exclusively on wastewater treatment infrastructure while neglecting upstream source reduction opportunities at pharmaceutical manufacturing facilities (Miarov et al. 2020). This fragmented regulatory design fails to internalize the externality costs that polluting enterprises impose on downstream treatment

operations, creating systematic misalignment between private incentives and social welfare objectives. Furthermore, the absence of PPCPs-specific effluent standards in most jurisdictions means that treatment plant operators lack clear compliance targets against which to evaluate technology investment decisions, while pharmaceutical manufacturers face no direct regulatory pressure to modify production processes that generate persistent micropollutants (Liu et al. 2024). The voluntary nature of many existing green manufacturing initiatives introduces collective action dilemmas wherein individual enterprises rationally defer costly process modifications while hoping competitors will bear the transition costs, resulting in suboptimal pollution reduction outcomes across the industry. Cross-jurisdictional regulatory heterogeneity compounds these challenges by enabling regulatory arbitrage, as pharmaceutical production may relocate to regions with less stringent environmental oversight, thereby displacing rather than eliminating contamination sources (El Hammoudani et al. 2024).

Understanding the strategic calculus of each stakeholder group proves essential for designing effective governance interventions. Pharmaceutical enterprises operate within competitive market environments where production cost minimization constitutes a primary operational objective, creating inherent tensions between environmental responsibility and profit maximization that shape clean production adoption decisions (Liang et al. 2023). These enterprises weigh the immediate costs of green manufacturing processes, including equipment upgrades, raw material substitutions, and process reengineering, against uncertain future benefits such as enhanced brand reputation, market access to environmentally conscious consumers, and reduced regulatory risk exposure. Wastewater treatment plants, whether publicly owned utilities or privately operated facilities, face capital constraints that limit technology upgrading capacity, particularly for AOPs and membrane separation systems requiring substantial initial investment and ongoing operational expenditures (Risch et al. 2022). Their technology adoption decisions respond to the interplay between regulatory compliance requirements, available fiscal support mechanisms, and the economic viability of reclaimed water utilization that could offset treatment costs. Government environmental protection agencies navigate complex political economy landscapes where regulatory stringency decisions must balance environmental protection mandates against industry competitiveness concerns, fiscal constraints on enforcement capacity, and accountability pressures that may favor visible short-term outcomes over sustained long-term environmental investments (Haoyang et al. 2022). The bounded rationality characterizing all three stakeholder groups means that strategic choices emerge through iterative learning and adaptation processes rather than instantaneous optimization, necessitating analytical frameworks capable of capturing such behavioral dynamics.

Public participation constitutes an increasingly salient dimension of environmental governance that shapes the strategic context within which enterprises, treatment facilities, and regulatory agencies operate. Household pharmaceutical disposal behaviors directly influence municipal wastewater PPCPs loading, while consumer-purchasing preferences can reward or

penalize manufacturers based on environmental performance credentials (Wang et al. 2015). Survey research documents significant public willingness to pay for PPCPs pollution control measures, indicating latent social demand for environmental protection that policymakers could potentially mobilize to support regulatory initiatives. Media attention to pharmaceutical contamination incidents amplifies reputational consequences for non-compliant enterprises and creates political pressure on government agencies to demonstrate regulatory effectiveness. While the present study focuses on the tripartite interactions among enterprises, treatment plants, and government agencies as the primary decision-makers in PPCPs governance, the broader social context within which these actors operate influences their strategic calculations and merits explicit acknowledgment as a boundary condition shaping model applicability.

Notwithstanding the extensive technical knowledge accumulated regarding PPCP occurrence, fate, and treatment, a critical dimension of micropollutant control remains underexplored: the governance mechanisms through which stakeholder behaviors are shaped and coordinated. Addressing PPCP pollution effectively requires concerted action among pharmaceutical manufacturers who generate contamination at source, wastewater treatment facilities charged with end-of-pipe removal, and government agencies possessing regulatory and fiscal instruments capable of influencing private sector decisions. Survey research among expert stakeholders drawn from government, academia, and industry across North America and Europe reveals broad consensus that pharmaceuticals in the environment constitute a legitimate concern for both ecosystem and human health, with respondents identifying advanced wastewater treatment, pharmaceutical return programs, and minimum secondary treatment requirements as promising management strategies (Doerr-MacEwen and Haight 2006). Public willingness to pay for measures aimed at reducing PPCP pollution has been documented in contingent valuation experiments, suggesting that societal demand for environmental protection exists and could potentially support policy interventions (Wang et al. 2015). Comparative analyses of regulatory frameworks across jurisdictions reveal substantial heterogeneity in approaches to pharmaceutical monitoring and standard-setting, with most countries lacking comprehensive regulatory oversight for these compounds in water resources (Miarov et al. 2020). This regulatory patchwork reflects underlying uncertainties regarding appropriate policy instruments and implementation strategies, uncertainties that cannot be resolved through technical analysis alone but require systematic examination of stakeholder incentives and behavioral dynamics.

The strategic interactions among actors in environmental governance systems can be productively analyzed using game-theoretic frameworks, which formalize how rational agents with potentially conflicting interests make interdependent decisions. Evolutionary game theory extends classical game theory by relaxing the assumption of perfect rationality, instead modeling strategy adoption as a population-level learning process wherein agents gradually shift toward behaviors that yield higher payoffs relative to alternatives. This approach proves particularly apt for environmental policy contexts where stakeholders operate under bounded rationality, possess incomplete information about others' strategies, and

adjust their decisions iteratively based on observed outcomes. Bibliometric analysis of micropollutant research demonstrates that technological innovation has dominated the field, with comparatively sparse investigation into policy implementation dynamics and stakeholder decision-making processes (El Hammoudani et al. 2024). The behavioral dimension of PPCP governance thus constitutes a notable gap; while we understand increasingly well how to remove these compounds technically, we understand far less about the conditions under which pharmaceutical enterprises will voluntarily adopt cleaner production practices, treatment plants will invest in technology upgrading, and government agencies will commit resources to regulatory enforcement.

This gap assumes particular significance because the decisions of these three stakeholder groups are inherently interdependent. Pharmaceutical enterprises choosing between clean production and traditional manufacturing impose differential pollution loads on downstream treatment infrastructure, affecting the costs and feasibility of achieving effluent quality targets. Wastewater treatment plants deciding whether to upgrade their technology do so within economic constraints shaped partly by influent characteristics determined by upstream enterprise behavior. Government agencies contemplating regulatory stringency must weigh enforcement costs against environmental benefits, with both influenced by the prevailing distribution of strategies among regulated entities. These feedback relationships create complex dynamics that single-actor analyses cannot capture; understanding system behavior requires examining how strategies co-evolve across all three populations simultaneously.

Against this background, several questions emerge as deserving systematic investigation. How do pharmaceutical enterprises, wastewater treatment plants, and government environmental protection agencies interact strategically within the PPCP pollution control system, and what behavioral dynamics arise from their interdependent decision-making? Under what combinations of cost structures, benefit distributions, and policy instrument intensities will stakeholders voluntarily converge toward environmentally beneficial strategies without regulatory compulsion? Conversely, under what conditions does sustained government intervention become necessary to induce and maintain cooperative behavior among private actors? Which policy levers exert the greatest influence on system trajectories, and how should these instruments be calibrated to maximize governance effectiveness?

This study addresses these questions through the development and analysis of a tripartite evolutionary game model capturing the strategic interactions among pharmaceutical enterprises, wastewater treatment plants, and government environmental agencies in the context of PPCP pollution control. The model specifies binary strategy spaces for each actor type, derives pay-off functions reflecting the economic incentives and cost structures characterizing real-world decision contexts, and employs replicator dynamics to trace the evolution of strategy frequencies within each population over time. Stability analysis of the resulting nonlinear dynamical system identifies equilibrium configurations and characterizes the parameter conditions under which each equilibrium emerges as asymptotically stable.

Numerical simulations validate theoretical predictions and explore sensitivity to key parameters.

The contributions of this research are threefold. First, this study extends the application of evolutionary game theory to the domain of PPCP governance, offering a novel analytical framework for examining stakeholder interactions in micropollutant control that complements the predominantly technical orientation of prior work. Second, the identification of stable equilibrium configurations and their associated parameter conditions provides actionable guidance for policy design, revealing circumstances under which market mechanisms suffice to achieve environmental objectives versus circumstances requiring sustained regulatory engagement. Third, sensitivity analysis pinpoints leverage points where policy interventions can most effectively redirect system dynamics toward socially desirable outcomes, informing the calibration of subsidy magnitudes, penalty intensities, and enforcement probabilities in practical regulatory contexts. These findings contribute to the theoretical understanding of environmental governance mechanisms while offering practical insights for addressing pharmaceutical pollution in aquatic environments, a challenge of growing urgency as global pharmaceutical consumption continues to expand and climate change intensifies pressures on water resources.

2 | Methodology

2.1 | Model Assumptions

The effective management of PPCPs in aquatic environments requires coordinated action among multiple stakeholders, each of whom possesses distinct interests, resource constraints, and strategic considerations. To systematically analyze the behavioral interactions and strategy evolution among these stakeholders during the adoption process of advanced treatment technologies, this study develops a tripartite evolutionary game model that captures the dynamic decision-making processes of pharmaceutical enterprises, wastewater treatment plants, and government environmental protection agencies. Because each participant operates under conditions of bounded rationality rather than perfect information, their strategic choices emerge through iterative learning and adaptation rather than instantaneous optimization, which necessitates an evolutionary game-theoretic approach that can accommodate such behavioral complexity.

The first assumption (H1) establishes the identity and role of each participant within the PPCPs pollution control system. Pharmaceutical enterprises constitute the primary point-source emitters of PPCPs, as their manufacturing processes inevitably generate wastewater streams containing active pharmaceutical ingredients, intermediates, and various organic compounds that exhibit persistence and potential ecotoxicological effects in receiving waters. Wastewater treatment plants serve as the critical end-of-pipe barriers responsible for removing these micropollutants before treated effluent enters natural water bodies, yet conventional biological treatment processes demonstrate limited efficacy against many pharmaceutical compounds due to their recalcitrant

molecular structures and low biodegradability. Government environmental protection agencies occupy the regulatory position within this tripartite system, wielding policy instruments such as emission standards, financial incentives, and enforcement mechanisms that can significantly influence the strategic calculus of both enterprises and treatment facilities. The interactions among these three bounded-rational groups give rise to complex behavioral dynamics that this evolutionary game model seeks to elucidate.

The second assumption (H2) specifies the binary strategy space available to each participant, which, while representing a simplification of real-world decision complexity, enables tractable mathematical analysis while preserving the essential strategic tensions inherent in PPCPs governance. Pharmaceutical enterprises face a choice between clean production strategies and traditional production strategies, where the former involves adopting green manufacturing processes, optimizing product formulations, implementing source reduction measures, and deploying on-site pretreatment systems to minimize PPCPs discharge into municipal sewerage systems, whereas the latter maintains conventional production methods that externalize pollution costs onto downstream treatment infrastructure and the broader environment. Wastewater treatment plants must decide between technology upgrading strategies and conventional treatment strategies, with the upgrading option requiring substantial capital investment in AOPs, granular activated carbon adsorption systems, membrane separation technologies, or hybrid treatment trains capable of achieving high removal efficiencies for diverse pharmaceutical compounds, while the conventional option relies on existing activated sludge processes that, despite their effectiveness for traditional pollutant parameters, achieve only incidental and often inadequate removal of many PPCPs. Government regulatory agencies choose between active regulation strategies and passive regulation strategies, where active regulation encompasses the promulgation of PPCPs-specific discharge limits, provision of technology upgrading subsidies, implementation of rigorous monitoring programs, and imposition of penalties for non-compliance, whereas passive regulation implies regulatory forbearance that shifts environmental and public health costs onto society while avoiding immediate fiscal expenditures on enforcement infrastructure.

The third assumption (H3) introduces the probability parameters that characterize the population-level distribution of strategies at any given time and whose temporal evolution the model seeks to track. Let x denote the probability that a representative pharmaceutical enterprise adopts clean production, with $1 - x$ representing the probability of traditional production; let y denote the probability that a representative wastewater treatment plant chooses technology upgrading, with $1 - y$ representing conventional treatment; and let z denote the probability that the government implements active regulation, with $1 - z$ corresponding to passive regulation. These probabilities satisfy the constraints $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$, and their dynamics over time reflect the aggregate outcome of individual learning processes as participants observe payoff differences and adjust their strategies accordingly. The complete set of model parameters and their economic interpretations are formally defined in Table 1.

2.2 | Game Strategies and Payoff Matrix

Building upon the assumptions and parameter definitions established above, this section constructs the complete payoff matrix that governs the evolutionary dynamics of the tripartite game system. The matrix encompasses all eight possible strategy combinations arising from the binary choices available to each participant, and careful specification of the payoff functions ensures that the model captures the essential economic incentives, cost structures, and externality relationships that shape stakeholder behavior in PPCPs governance contexts. Each cell of the payoff matrix reflects the net benefit or loss that each participant realizes under a specific strategy profile, accounting for direct revenues and costs as well as the interactive effects that one party's choices impose on others. The complete payoff structure for all eight strategy combinations is presented in Table 2.

Consider first the strategy combination in which pharmaceutical enterprises adopt clean production, wastewater treatment plants implement technology upgrading, and government pursues active regulation. Under this configuration, pharmaceutical enterprises realize revenue R_1 from their environmentally responsible market positioning while incurring clean production costs C_1 and receiving government subsidies S_1 , yielding a net payoff of $R_1 - C_1 + S_1$. Wastewater treatment plants obtain technology upgrading revenue R_3 , bear the substantial capital and operational costs C_3 associated with advanced treatment systems, and receive government financial support S_2 , producing a net payoff of $R_3 - C_3 + S_2$. Government captures the societal benefits R_5 from effective PPCPs control, expends regulatory resources C_6 on monitoring and enforcement, and disburses subsidies S_1 and S_2 to incentivize desirable behavior, resulting in a net payoff of $R_5 - C_6 - S_1 - S_2$. The positive-sum character of this outcome, wherein all parties can potentially benefit from cooperative behavior when supported by appropriate policy instruments, represents the socially optimal equilibrium toward which policy interventions should aim to steer system evolution.

When pharmaceutical enterprises maintain clean production and wastewater treatment plants upgrade their technology but government adopts passive regulation, the absence of both subsidies and penalties alters the payoff structure considerably. Pharmaceutical enterprises still earn R_1 and pay C_1 but receive no subsidy, yielding $R_1 - C_1$; wastewater treatment plants similarly obtain $R_3 - C_3$ without government financial support; and government enjoys the environmental benefits R_5 without incurring regulatory costs, which might appear advantageous in the short term but ignores the fragility of voluntary compliance in the absence of enforcement credibility. The situation changes markedly when pharmaceutical enterprises choose clean production and government regulates actively, but wastewater treatment plants opt for conventional treatment, in which case the treatment facilities face penalties F_2 and reputational damages L_2 for failing to meet effluent PPCPs standards despite receiving influent with relatively low pharmaceutical loading, producing a payoff of $R_4 - C_4 - F_2 - L_2$ that creates strong incentives for technology adoption.

The payoff implications become particularly consequential when pharmaceutical enterprises select traditional production,

TABLE 1 | Model parameters and definitions.

Parameter	Definition
R_1	Revenue accruing to pharmaceutical enterprises from clean production, including green brand premiums, enhanced market competitiveness, and improved corporate social responsibility reputation
R_2	Revenue accruing to pharmaceutical enterprises from traditional production
R_3	Revenue accruing to wastewater treatment plants from technology upgrading, encompassing improved effluent quality, reclaimed water utilization benefits, and enhanced social reputation
R_4	Baseline operational revenue accruing to wastewater treatment plants from conventional treatment
R_5	Potential benefits accruing to government from active regulation, including public health protection, ecological environment improvement, and enhanced governance performance
C_1	Costs incurred by pharmaceutical enterprises for clean production implementation, including process modification, green raw material substitution, and on-site pretreatment investment
C_2	Costs incurred by pharmaceutical enterprises for traditional production
C_3	Costs incurred by wastewater treatment plants for technology upgrading, encompassing capital investment and operational expenditure for AOPs, activated carbon adsorption, and membrane technologies
C_4	Operational costs incurred by wastewater treatment plants for conventional treatment
C_5	Additional treatment costs imposed on wastewater treatment plants when pharmaceutical enterprises employ traditional production, arising from elevated influent PPCPs concentrations
C_6	Costs incurred by government for active regulation implementation, including monitoring equipment, enforcement personnel, and analytical testing expenses
C_7	Environmental health damages and social costs borne by government under passive regulation
S_1	Subsidies provided by government to pharmaceutical enterprises adopting clean production under active regulation
S_2	Subsidies provided by government to wastewater treatment plants undertaking technology upgrading under active regulation
F_1	Fines imposed by government on pharmaceutical enterprises maintaining traditional production under active regulation
F_2	Fines imposed by government on wastewater treatment plants with non-compliant effluent PPCPs concentrations under active regulation
L_1	Reputational losses suffered by pharmaceutical enterprises from regulatory penalties for traditional production
L_2	Reputational losses suffered by wastewater treatment plants from effluent quality noncompliance

as this choice imposes negative externalities on downstream wastewater treatment infrastructure regardless of the treatment strategy employed. Should treatment plants upgrade their technology while enterprises pollute and government regulates, the plants must absorb additional treatment costs C_5 attributable to elevated influent PPCPs concentrations even as they receive subsidies S_2 , yielding $R_3 - C_3 - C_5 + S_2$, while enterprises suffer fines F_1 and reputational losses L_1 that reduce their payoff to $R_2 - C_2 - F_1 - L_1$. The most problematic scenario arises when all three parties defect from socially beneficial strategies: enterprises pursue traditional production, treatment plants maintain conventional processes, and government remains passive, whereupon enterprises capture $R_2 - C_2$ without penalty, treatment plants obtain $R_4 - C_4 - C_5$ as they struggle with high PPCPs loading using inadequate technology, and government bears environmental damage costs $-C_7$ that ultimately manifest as degraded water quality, ecosystem impairment, and public health risks.

3 | Replicator Dynamics and Equilibrium Analysis

3.1 | Replicator Dynamic Equations

The replicator dynamics framework provides the mathematical apparatus for analyzing how strategy frequencies evolve within each population as participants observe realized payoffs and gradually shift toward more successful behavioral patterns. An evolutionarily stable strategy (ESS) represents a strategy configuration that, once established in a population, resists invasion by alternative strategies because any small group of mutants adopting different behavior would obtain lower average payoffs than the incumbent population. Deriving the replicator dynamic equations for each party requires computing expected payoffs under each pure strategy, calculating the population average payoff, and specifying the rate of strategy adjustment as proportional to the payoff differential, which captures the intuition that strategies outperforming the

TABLE 2 | Strategy combinations and payoff matrix for three-party evolutionary game.

Strategy combination	Pharmaceutical enterprises	Wastewater treatment plants	Government environmental agency
(Clean production, technology upgrading, active regulation)	$R_1 - C_1 + S_1$	$R_3 - C_3 + S_2$	$R_5 - C_6 - S_1 - S_2$
(Clean production, technology upgrading, passive regulation)	$R_1 - C_1$	$R_3 - C_3$	R_5
(Clean production, conventional treatment, active regulation)	$R_1 - C_1 + S_1$	$R_4 - C_4 - F_2 - L_2$	$R_5 - C_6 - S_1 + F_2$
(Clean production, conventional treatment, Passive regulation)	$R_1 - C_1$	$R_4 - C_4$	$R_5 - C_7$
(Traditional production, technology upgrading, active regulation)	$R_2 - C_2 - F_1 - L_1$	$R_3 - C_3 - C_5 + S_2$	$-C_6 - S_2 + F_1$
(Traditional production, technology upgrading, passive regulation)	$R_2 - C_2$	$R_3 - C_3 - C_5$	$-C_7$
(Traditional production, conventional treatment, active regulation)	$R_2 - C_2 - F_1 - L_1$	$R_4 - C_4 - C_5 - F_2 - L_2$	$-C_6 + F_1 + F_2$
(Traditional production, conventional treatment, passive regulation)	$R_2 - C_2$	$R_4 - C_4 - C_5$	$-C_7$

population average will spread while underperforming strategies will decline.

For pharmaceutical enterprises, the expected payoff from clean production U_{x1} aggregates the payoffs realized across all possible strategy combinations of the other two parties, weighted by the respective probabilities of those combinations occurring. When treatment plants upgrade with probability y and government regulates actively with probability z , enterprises choosing clean production obtain the highest payoff component $(R_1 - C_1 + S_1)$ with probability yz ; when treatment plants upgrade but government remains passive, the payoff $(R_1 - C_1)$ occurs with probability $y(1 - z)$; analogous reasoning applies to the remaining combinations, yielding:

$$U_{x1} = (R_1 - C_1 + S_1)yz + (R_1 - C_1)y(1 - z) + (R_1 - C_1 + S_1)(1 - y)z + (R_1 - C_1)(1 - y)(1 - z)$$

Algebraic simplification reveals that the wastewater treatment plant's strategy choice does not directly affect the pharmaceutical enterprise's payoff from clean production, because clean production eliminates the externality channel through which enterprise behavior influences treatment costs, so the expected payoff reduces to:

$$U_{x1} = R_1 - C_1 + S_1z$$

The expected payoff from traditional production U_{x2} follows similar logic but incorporates the penalties and reputational losses that enterprises suffer when government regulates actively:

$$U_{x2} = (R_2 - C_2 - F_1 - L_1)yz + (R_2 - C_2)y(1 - z) + (R_2 - C_2 - F_1 - L_1)(1 - y)z + (R_2 - C_2)(1 - y)(1 - z)$$

This expression simplifies to:

$$U_{x2} = R_2 - C_2 - (F_1 + L_1)z$$

The average expected payoff for the enterprise population equals $\bar{U}_x = xU_{x1} + (1 - x)U_{x2}$, and the replicator dynamic equation governing the evolution of clean production adoption takes the standard form:

$$H(x) = \frac{dx}{dt} = x(U_{x1} - \bar{U}_x) = x(1 - x)(U_{x1} - U_{x2})$$

Substituting the simplified expected payoff expressions yields the enterprise replicator dynamics:

$$H(x) = x(1 - x)[(R_1 - C_1) - (R_2 - C_2) + (S_1 + F_1 + L_1)z]$$

This equation reveals that the direction of enterprise strategy evolution depends critically on the comparison between the net benefit differential of clean versus traditional production $(R_1 - C_1) - (R_2 - C_2)$ and the policy-induced incentive term $(S_1 + F_1 + L_1)z$ that government regulation activates. When the intrinsic profitability advantage of traditional production exceeds the expected regulatory penalty, enterprises will drift toward polluting behavior unless government increases either the probability or intensity of enforcement.

For wastewater treatment plants, the expected payoff from technology upgrading U_{y1} must account for the additional treatment burden C_5 imposed when upstream pharmaceutical enterprises employ traditional production:

$$U_{y1} = (R_3 - C_3 + S_2)xz + (R_3 - C_3)x(1 - z) + (R_3 - C_3 - C_5 + S_2)(1 - x)z + (R_3 - C_3 - C_5)(1 - x)(1 - z)$$

Simplification yields:

$$U_{y1} = R_3 - C_3 - C_5(1-x) + S_2z$$

The expected payoff from conventional treatment U_{y2} incorporates both the elevated treatment costs from enterprise pollution and the penalties for effluent non-compliance under active regulation:

$$U_{y2} = (R_4 - C_4 - F_2 - L_2)xz + (R_4 - C_4)x(1-z) + (R_4 - C_4 - C_5 - F_2 - L_2)(1-x)z + (R_4 - C_4 - C_5)(1-x)(1-z)$$

Which reduces to:

$$U_{y2} = R_4 - C_4 - C_5(1-x) - (F_2 + L_2)z$$

The treatment plant replicator dynamics follow as:

$$H(y) = \frac{dy}{dt} = y(1-y)(U_{y1} - U_{y2})$$

$$H(y) = y(1-y)[(R_3 - C_3) - (R_4 - C_4) + (S_2 + F_2 + L_2)z]$$

Notably, the enterprise clean production probability x affects both U_{y1} and U_{y2} equally through the externality cost term $C_5(1-x)$, which cancels in the payoff differential and therefore does not appear in the simplified replicator equation. This mathematical result merits careful interpretation rather than dismissal as a modeling limitation. The cancellation reflects a substantive economic reality: both technology upgrading and conventional treatment options face identical baseline cost increases when upstream pharmaceutical enterprises employ traditional production methods, because elevated influent PPCPs concentrations impose additional operational burdens regardless of the treatment technology employed. The critical distinction lies not in differential exposure to upstream pollution costs, but in the capacity of upgraded technologies to achieve regulatory compliance and capture associated benefits including government subsidies, avoided penalties, and enhanced social reputation. This finding carries important policy implications: market-based coordination between upstream and downstream actors cannot spontaneously emerge through cost transmission alone, necessitating government intervention through subsidy and penalty instruments to create the incentive structures that induce treatment plant technology adoption. The model thus reveals that PPCPs governance systems exhibit a form of strategic independence at the private sector level that only regulatory oversight can overcome, a conclusion that strengthens rather than undermines the case for active government engagement in micropollutant control. This mathematical result carries the important policy implication that upstream source control, while reducing absolute treatment costs, does not directly alter the relative attractiveness of technology upgrading versus conventional treatment; instead, government policy instruments $(S_2 + F_2 + L_2)z$ drive the treatment plant's technology adoption decision.

For government environmental protection agencies, the expected payoff from active regulation U_{z1} reflects the heterogeneous

outcomes that arise depending on enterprise and treatment plant behavior:

$$U_{z1} = (R_5 - C_6 - S_1 - S_2)xy + (R_5 - C_6 - S_1 + F_2)x(1-y) + (-C_6 - S_2 + F_1)(1-x)y + (-C_6 + F_1 + F_2)(1-x)(1-y)$$

The expected payoff from passive regulation U_{z2} captures the environmental benefits when enterprises comply voluntarily and the social costs when they do not:

$$U_{z2} = R_5xy + (R_5 - C_7)x(1-y) + (-C_7)(1-x)y + (-C_7)(1-x)(1-y)$$

Which simplifies to:

$$U_{z2} = R_5x - C_7(1-x)$$

The government replicator dynamics take the form:

$$H(z) = \frac{dz}{dt} = z(1-z)(U_{z1} - U_{z2})$$

The complexity of U_{z1} precludes simple closed-form simplification, but the essential insight remains that government's optimal regulatory intensity depends on the current distribution of strategies in the enterprise and treatment plant populations, creating feedback loops that can generate rich evolutionary dynamics including limit cycles and path-dependent equilibrium selection.

3.2 | Tripartite Evolutionary Game Replicator Dynamic System

Consolidating the replicator dynamic equations derived above produces the coupled nonlinear differential equation system that governs the co-evolution of strategies across all three populations:

$$\begin{aligned} \frac{dx}{dt} &= x(1-x)[(R_1 - C_1) - (R_2 - C_2) + (S_1 + F_1 + L_1)z] \\ \frac{dy}{dt} &= y(1-y)[(R_3 - C_3) - (R_4 - C_4) + (S_2 + F_2 + L_2)z] \\ \frac{dz}{dt} &= z(1-z)(U_{z1} - U_{z2}) \end{aligned}$$

This system of equations encapsulates the interdependent evolutionary trajectories of clean production adoption among pharmaceutical enterprises, technology upgrading among wastewater treatment plants, and regulatory activism among government agencies. The nonlinear coupling among the three equations implies that small perturbations in one population's strategy distribution can propagate through the system to influence equilibrium outcomes for all parties, which underscores the importance of coordinated policy design that accounts for these cross-population feedback effects. Subsequent analysis will identify the equilibrium points of this system by setting all three time derivatives equal to zero, assess the local stability properties of each equilibrium through linearization and eigenvalue analysis of the Jacobian matrix, and employ numerical

simulation to explore how variations in key parameters affect the basins of attraction and evolutionary paths connecting initial conditions to long-run outcomes.

4 | Stability Analysis of Equilibrium Points

4.1 | Jacobian Matrix and Eigenvalue Computation

To determine the local stability properties of equilibrium points within the tripartite evolutionary game system, this study employs the Jacobian matrix approach, which involves linearizing the nonlinear replicator dynamic system around each equilibrium and examining the signs of the resulting eigenvalues. At pure strategy equilibrium points where each probability variable assumes a boundary value of either 0 or 1, the Jacobian matrix becomes diagonal because cross-partial derivative terms containing products such as $x(1-x)$, $y(1-y)$, and $z(1-z)$ evaluate to zero. This diagonalization property allows the eigenvalues to be extracted directly from the diagonal entries, thereby circumventing the computational complexity associated with characteristic polynomial factorization.

Table 3 summarizes the eigenvalue expressions evaluated at each of the eight pure strategy equilibria. Evaluating the Jacobian matrix at each of the eight pure strategy equilibria yields the eigenvalue expressions summarized in the following table:

4.2 | Rationale for Excluding Mixed Strategy Equilibria

This analysis restricts attention to pure strategy equilibria for three reasons. First, pure strategy configurations possess

direct practical interpretability because they correspond to scenarios where all members of a stakeholder group adopt identical behavioral patterns, whereas mixed equilibria would require individual actors to randomize decisions according to specific probability weights. Second, mixed strategy equilibria in asymmetric multipopulational evolutionary games typically exhibit structural instability, meaning arbitrarily small parameter perturbations can eliminate such equilibria entirely. Third, interior equilibria generically function as saddle points rather than attractors in tripartite game systems, failing to satisfy asymptotic stability conditions even when they exist mathematically.

4.3 | Stability Conditions and Impossibility Results

An equilibrium achieves asymptotic stability if and only if all three eigenvalues possess strictly negative real parts. Systematic examination of the eigenvalue expressions reveals distinct stability requirements for each equilibrium, along with two configurations that cannot achieve stability under any parameter specification.

E1(0,0,0) achieves stability when $(R_2 - C_2) > (R_1 - C_1)$, $(R_4 - C_4) > (R_3 - C_3)$, and $(C_6 - F_1 - F_2) > C_7$. These conditions characterize a scenario where traditional production and conventional treatment generate higher net profits than their environmentally beneficial alternatives, while government perceives regulatory implementation costs as exceeding the sum of environmental damage costs and potential fine revenues.

E2(1,0,0) achieves stability when $(R_1 - C_1) > (R_2 - C_2)$, $(R_4 - C_4) > (R_3 - C_3)$, and $(C_6 + S_1 - F_2) > C_7$. Clean production must be intrinsically profitable, technology upgrading must

TABLE 3 | Eigenvalues of Jacobian matrix at pure strategy equilibria.

Equilibrium	λ_1	λ_2	λ_3
E1(0,0,0)	$(R_1 - C_1) - (R_2 - C_2)$	$(R_3 - C_3) - (R_4 - C_4)$	$(C_7 - C_6) + F_1 + F_2$
E2(1,0,0)	$(R_2 - C_2) - (R_1 - C_1)$	$(R_3 - C_3) - (R_4 - C_4)$	$(C_7 - C_6) + F_2 - S_1$
E3(0,1,0)	$(R_1 - C_1) - (R_2 - C_2)$	$(R_4 - C_4) - (R_3 - C_3)$	$(C_7 - C_6) + F_1 - S_2$
E4(0,0,1)	$(R_1 - C_1) - (R_2 - C_2) + (S_1 + F_1 + L_1)$	$(R_3 - C_3) - (R_4 - C_4) + (S_2 + F_2 + L_2)$	$(C_6 - C_7) - F_1 - F_2$
E5(1,1,0)	$(R_2 - C_2) - (R_1 - C_1)$	$(R_4 - C_4) - (R_3 - C_3)$	$-C_6 - S_1 - S_2$
E6(1,0,1)	$(R_2 - C_2) - (R_1 - C_1) - (S_1 + F_1 + L_1)$	$(R_3 - C_3) - (R_4 - C_4) + (S_2 + F_2 + L_2)$	$(C_6 - C_7) - F_2 + S_1$
E7(0,1,1)	$(R_1 - C_1) - (R_2 - C_2) + (S_1 + F_1 + L_1)$	$(R_4 - C_4) - (R_3 - C_3) - (S_2 + F_2 + L_2)$	$(C_6 - C_7) - F_1 + S_2$
E8(1,1,1)	$(R_2 - C_2) - (R_1 - C_1) - (S_1 + F_1 + L_1)$	$(R_4 - C_4) - (R_3 - C_3) - (S_2 + F_2 + L_2)$	$C_6 + S_1 + S_2$

remain economically unattractive, and government must find passive regulation preferable even after forgoing enterprise subsidy expenditures.

E3(0,1,0) achieves stability when $(R_2 - C_2) > (R_1 - C_1)$, $(R_3 - C_3) > (R_4 - C_4)$, and $(C_6 + S_2 - F_1) > C_7$. Technology upgrading must generate positive net returns while clean production remains unprofitable, with government preferring regulatory passivity.

E4(0,0,1) requires that $\lambda_1 < 0$ and $\lambda_2 < 0$ hold simultaneously with $\lambda_3 < 0$. The first two conditions demand that policy instruments fail to induce private sector cooperation, while the third requires government to find active regulation worthwhile. These requirements prove mutually inconsistent: if subsidies and penalties cannot shift enterprise and treatment plant behavior, the fiscal burden of regulation lacks justification. E4 therefore cannot constitute a stable equilibrium.

E5(1,1,0) has $\lambda_3 = -C_6 - S_1 - S_2$, which is automatically negative since regulatory costs and subsidy amounts are positive by definition. Stability additionally requires $(R_1 - C_1) > (R_2 - C_2)$ and $(R_3 - C_3) > (R_4 - C_4)$, meaning both clean production and technology upgrading must be intrinsically profitable without regulatory support.

E6(1,0,1) achieves stability when regulatory instruments effectively maintain enterprise compliance but fail to induce treatment plant upgrading, requiring $(R_1 - C_1) + (S_1 + F_1 + L_1) > (R_2 - C_2)$, $(R_4 - C_4) > (R_3 - C_3) + (S_2 + F_2 + L_2)$, and $(C_7 + F_2) > (C_6 + S_1)$.

E7(0,1,1) achieves stability under the converse conditions where regulatory pressure induces treatment plant upgrading but fails to shift enterprise behavior, requiring $R_2 - (C_2) > (R_1 - C_1) + (S_1 + F_1 + L_1)$, $(R_3 - C_3) + (S_2 + F_2 + L_2) > (R_4 - C_4)$, and $(C_7 + F_1) > (C_6 + S_2)$.

E8(1,1,1) has $\lambda_3 = C_6 + S_1 + S_2$, which is necessarily positive because regulatory costs and subsidy expenditures are positive quantities. When private actors already exhibit full cooperation, government faces fiscal incentives to discontinue costly regulatory programs, as environmental benefits accrue regardless of enforcement effort. E8 therefore cannot be asymptotically stable, implying that sustaining universal cooperation requires ongoing regulatory commitment to maintain enforcement credibility rather than representing a self-sustaining equilibrium configuration.

5 | Numerical Simulation and Stability Verification

5.1 | Parameter Configuration and Simulation Design

To validate the theoretical stability conditions derived from eigenvalue analysis and to visualize the evolutionary dynamics of the tripartite game system, this section presents numerical simulations for each potentially stable equilibrium configuration. The numerical simulations employ stylized

parameter values designed to satisfy the stability conditions derived from theoretical analysis and to illustrate the qualitative dynamic behaviors predicted by the model. Following established conventions in evolutionary game theory literature, parameter magnitudes are assigned to preserve economically meaningful relationships among cost, benefit, and policy instrument variables rather than to replicate specific empirical contexts. The ratio of clean production costs to traditional production costs reflects documented ranges for green manufacturing process adoption in pharmaceutical industries, while subsidy and penalty magnitudes correspond to typical policy instrument intensities observed in environmental governance programs (Haoyang et al. 2022; Liang et al. 2023). Life cycle assessment studies of advanced wastewater treatment technologies report that ozonation and granular activated carbon systems incur capital costs approximately 2–3 times higher than conventional activated sludge processes, with operational expenditures elevated by factors of 1.5–2.5 depending on target removal efficiencies and influent characteristics (Amaya-Santos et al. 2025; Rashid et al. 2023; Risch et al. 2022). These empirical ratios inform the specification of C_3 relative to C_4 in the simulation scenarios. The primary purpose of simulation analysis is to validate theoretical stability predictions and to explore how parameter variations influence convergence trajectories and equilibrium selection, providing qualitative guidance for policy design rather than quantitative forecasting of specific governance outcomes. The parameter values assigned to each scenario satisfy the corresponding stability conditions identified in Section 4.3, thereby ensuring that the simulated trajectories converge to the predicted equilibrium points. All simulations employ identical initial conditions with $x_0 = 0.2$, $y_0 = 0.3$, and $z_0 = 0.4$, representing a baseline scenario where pharmaceutical enterprises exhibit low clean production adoption, wastewater treatment plants demonstrate modest technology upgrading propensity, and government maintains moderate regulatory engagement. The simulation horizon extends to $T = 100$ time units, which proves sufficient for observing convergence behavior across all parameter configurations examined.

5.2 | Scenario Analysis for Stable Equilibria

The first simulation scenario examines conditions under which the system converges to the universal defection equilibrium E1(0,0,0), where pharmaceutical enterprises maintain traditional production, wastewater treatment plants operate conventional treatment processes, and government adopts passive regulation. The parameter configuration establishes $(R_2 - C_2) = 70 > (R_1 - C_1) = 30$, $(R_4 - C_4) = 50 > (R_3 - C_3) = 10$, and $(C_6 - F_1 - F_2) = 30 > C_7 = 20$, which collectively satisfy all three stability conditions for E1. These values reflect a market environment where traditional production yields substantially higher net profits than clean production due to elevated green technology costs, where conventional treatment remains economically attractive relative to capital-intensive advanced oxidation systems, and where government perceives regulatory implementation costs as exceeding environmental damage costs. Figure 1 displays the temporal evolution and three-dimensional phase trajectory for this scenario. The left panel reveals that all three strategy probabilities decline monotonically from their

initial values toward zero, with the enterprise clean production probability x and treatment plant upgrading probability y approaching the lower boundary within approximately 20-time units, while the government active regulation probability z exhibits slightly slower convergence. The right panel presents the corresponding stem plot in (x, y, z) phase space, illustrating how the system trajectory spirals downward from the initial condition $(0.2, 0.3, 0.4)$ toward the origin. This convergence pattern confirms that when environmentally beneficial strategies impose net costs on private actors and regulatory intervention appears fiscally unjustifiable, the evolutionary dynamics inevitably drive the system toward complete environmental governance failure.

The second scenario investigates the equilibrium configuration $E2(1, 0, 0)$, characterized by universal enterprise clean production adoption coupled with treatment plant and government defection. Parameter values satisfy $(R_1 - C_1) = 75 > (R_2 - C_2) = 50$, $(R_4 - C_4) = 45 > (R_3 - C_3) = 0$, and $(C_6 + S_1 - F_2) = 45 > C_7 = 25$. This configuration represents market conditions where green

brand premiums and consumer environmental preferences render clean production intrinsically profitable, yet advanced treatment technologies remain economically unattractive due to high capital requirements, and government finds passive regulation optimal because enterprise compliance occurs voluntarily without regulatory expenditure. Figure 2 demonstrates that the enterprise strategy probability x increases rapidly from 0.2 toward unity while both y and z decline toward zero. The three-dimensional trajectory in the right panel traces a path from the initial condition toward the point $(1, 0, 0)$ on the positive x -axis, confirming theoretical predictions. This equilibrium possesses practical significance because it suggests that sufficiently strong market incentives for green products can induce enterprise compliance without government intervention, although the failure of treatment plant upgrading implies that end-of-pipe PPCPs removal remains inadequate even when source pollution decreases.

The third scenario examines the configuration $E3(0, 1, 0)$, where treatment plants universally upgrade their

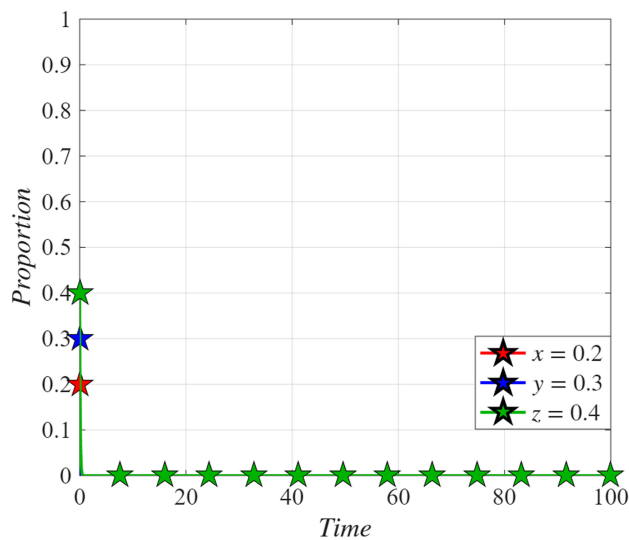


FIGURE 1 | Evolutionary dynamics converging to $E1(0,0,0)$.

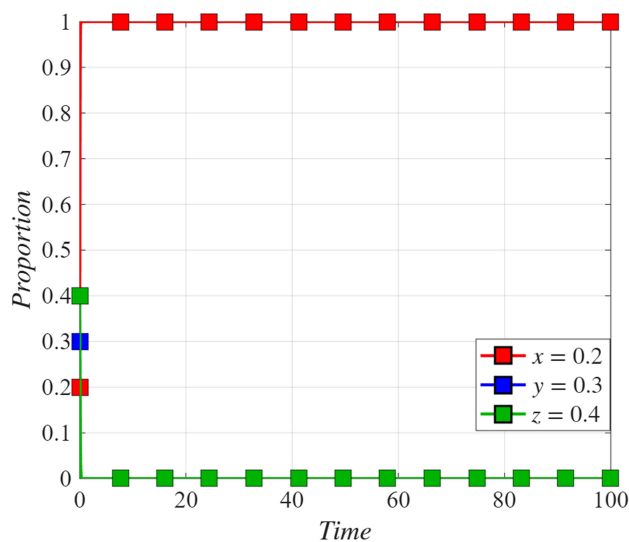
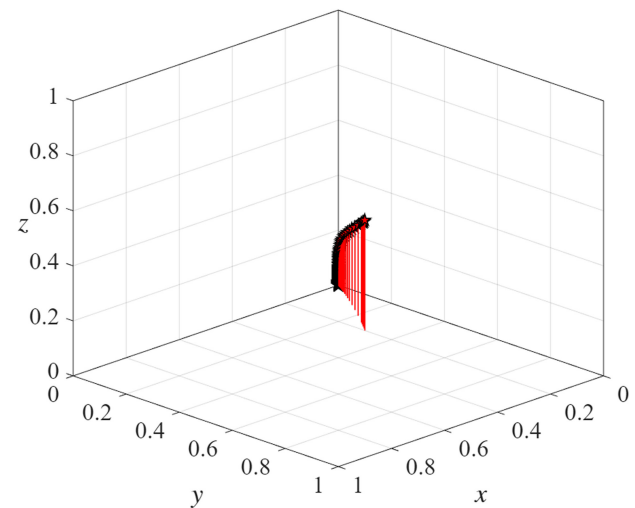
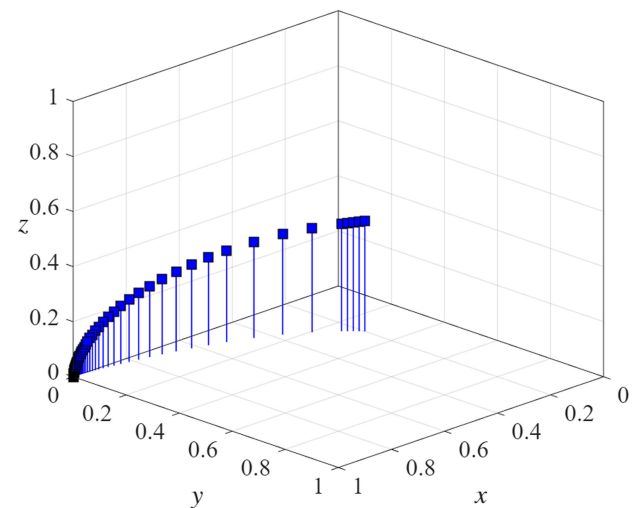


FIGURE 2 | Evolutionary dynamics converging to $E2(1,0,0)$.



technology while enterprises and government select non-cooperative strategies. The stability conditions require $(R_2 - C_2) = 70 > (R_1 - C_1) = 20$, $(R_3 - C_3) = 55 > (R_4 - C_4) = 30$, and $(C_6 + S_2 - F_1) = 51 > C_7 = 22$. Such conditions might arise in jurisdictions where reclaimed water markets create strong revenue incentives for treatment plants to improve effluent quality, where clean production imposes prohibitive costs on pharmaceutical manufacturers, and where government lacks political will or fiscal capacity for regulatory enforcement. The simulation results in Figure 3 confirm that y converges to unity while x and z approach zero. The phase trajectory demonstrates movement toward the point $(0, 1, 0)$ along the positive y -axis. Although treatment plant upgrading can achieve partial PPCPs removal, the persistence of traditional pharmaceutical production means that influent concentrations remain elevated, imposing additional treatment costs C_5 on upgraded facilities and potentially compromising removal efficiency for the most recalcitrant compounds.

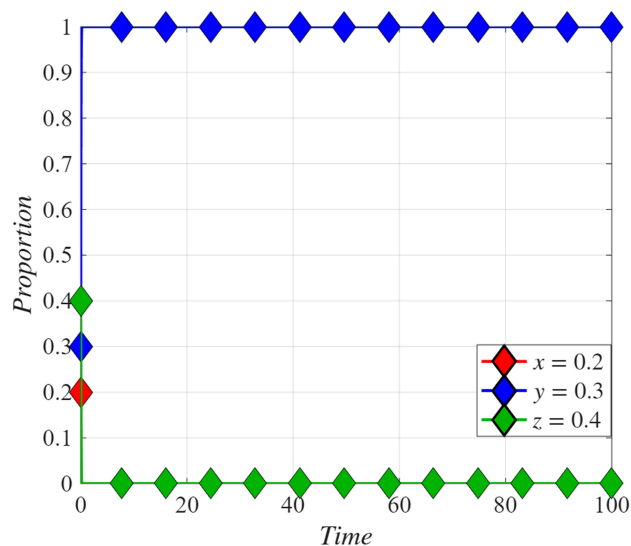


FIGURE 3 | Evolutionary dynamics converging to E3(0, 1, 0).

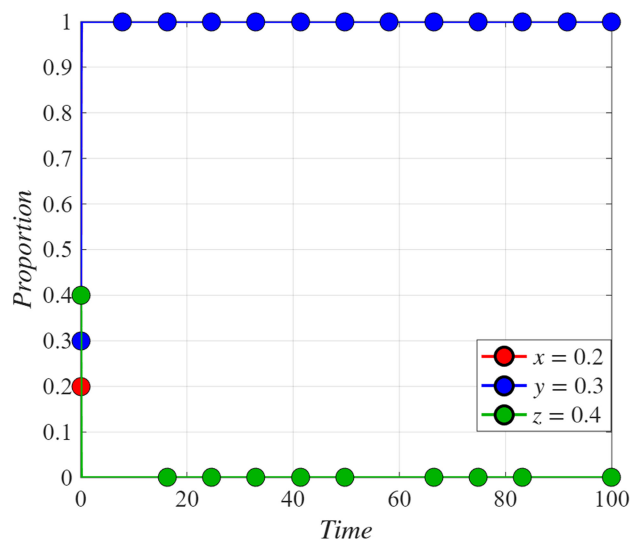
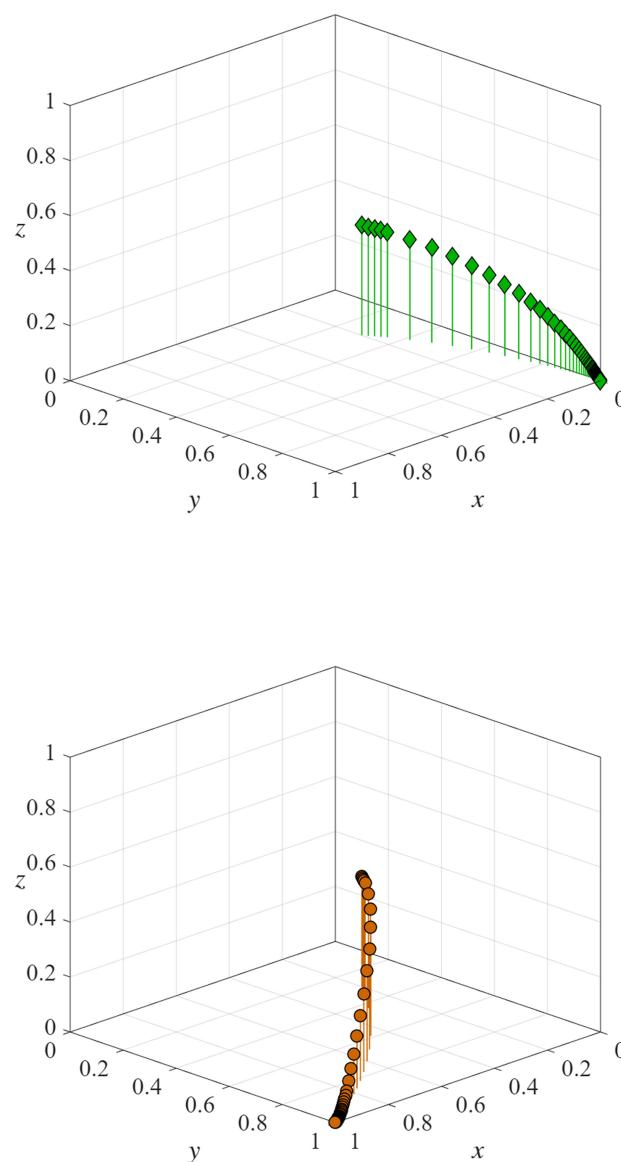


FIGURE 4 | Evolutionary dynamics converging to E5(1, 1, 0).

The fourth scenario represents arguably the most desirable stable configuration achievable within the model framework: voluntary private sector cooperation at E5(1, 1, 0), where both enterprises and treatment plants adopt environmentally beneficial strategies without government regulatory pressure. Stability requires $(R_1 - C_1) = 80 > (R_2 - C_2) = 50$ and $(R_3 - C_3) = 55 > (R_4 - C_4) = 25$, ensuring that clean production and technology upgrading are intrinsically profitable. The third eigenvalue $\lambda_3 = -C_6 - S_1 - S_2 = -70$ is automatically negative, guaranteeing government convergence toward passive regulation once private cooperation becomes established. Figure 4 illustrates the convergence dynamics toward E5. Both x and y increase rapidly toward unity as enterprises and treatment plants recognize the profitability advantages of their cooperative strategies, while z declines toward zero as government withdraws from costly regulatory activities that have become redundant. The stem plot trajectory moves toward the corner $(1, 1, 0)$ of the unit cube, representing full private sector cooperation with



government passivity. This equilibrium demonstrates that market-based mechanisms can achieve environmental objectives when technological progress or consumer preferences align private incentives with social welfare, although its stability depends critically on maintaining the profitability conditions that currently favor green strategies.

The fifth scenario explores the partial cooperation equilibrium $E6(1, 0, 1)$, where enterprises comply with clean production requirements under active government regulation, but treatment plants maintain conventional processes. The parameter configuration satisfies $(R_1 - C_1) + (S_1 + F_1 + L_1) = 125 > (R_2 - C_2) = 55$, $(R_4 - C_4) = 40 > (R_3 - C_3) + (S_2 + F_2 + L_2) = 28$, and $(C_7 + F_2) = 70 > (C_6 + S_1) = 55$. These values characterize situations where regulatory penalties and subsidies successfully induce enterprise compliance, yet technology upgrading costs remain prohibitively high for treatment plants even with government support, and environmental damages justify continued regulatory engagement. Figure 5 shows that x and z converge to unity while y approaches zero. The phase trajectory terminates at the edge $(1, 0, 1)$ of the

unit cube, representing an equilibrium where enterprises practice clean production under government oversight but treatment plants lack sufficient economic motivation to upgrade despite regulatory pressure. This configuration may reflect realistic constraints in developing economies where pharmaceutical manufacturers can feasibly adopt cleaner processes but municipal wastewater infrastructure faces severe capital constraints.

The sixth stable configuration examined is $E7(0, 1, 1)$, representing the mirror image of E 6 wherein treatment plants cooperate with government while enterprises defect. Stability conditions require $(R_2 - C_2) = 70 > (R_1 - C_1) + (S_1 + F_1 + L_1) = 43$, $(R_3 - C_3) + (S_2 + F_2 + L_2) = 125 > (R_4 - C_4) = 25$, and $(C_7 + F_1) = 65 > (C_6 + S_2) = 60$. This scenario might characterize industries where pharmaceutical production processes exhibit limited flexibility for green transformation, where treatment plant upgrading generates substantial ancillary benefits through reclaimed water utilization, and where government prioritizes end-of-pipe control over source reduction. Figure 6 confirms convergence to E7, with y and z increasing toward unity while x declines toward zero. The trajectory

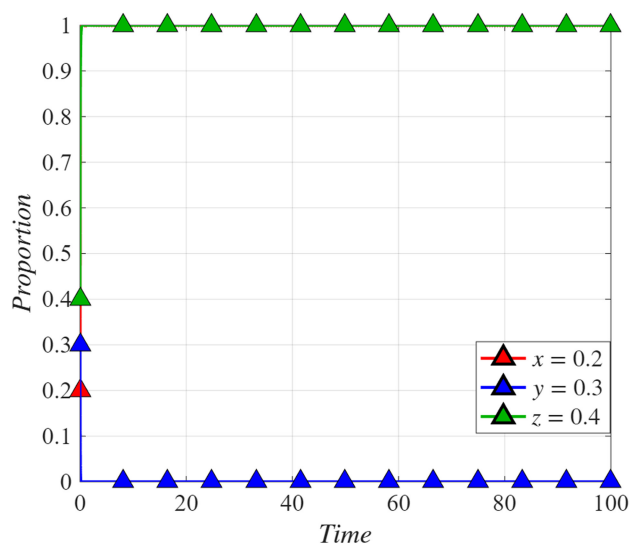


FIGURE 5 | Evolutionary dynamics converging to $E6(1, 0, 1)$.

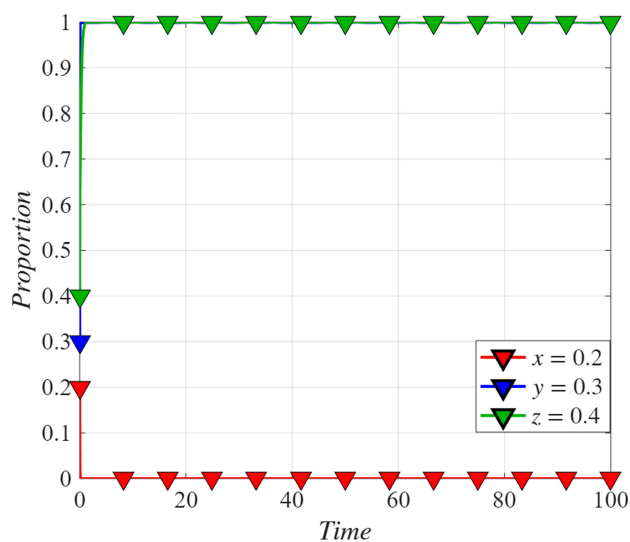
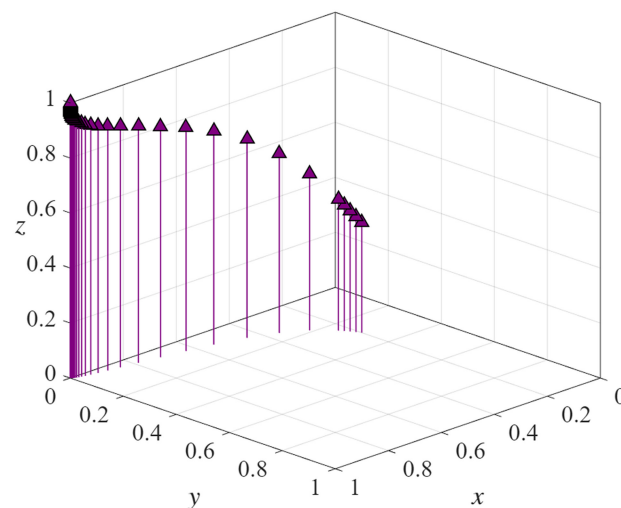
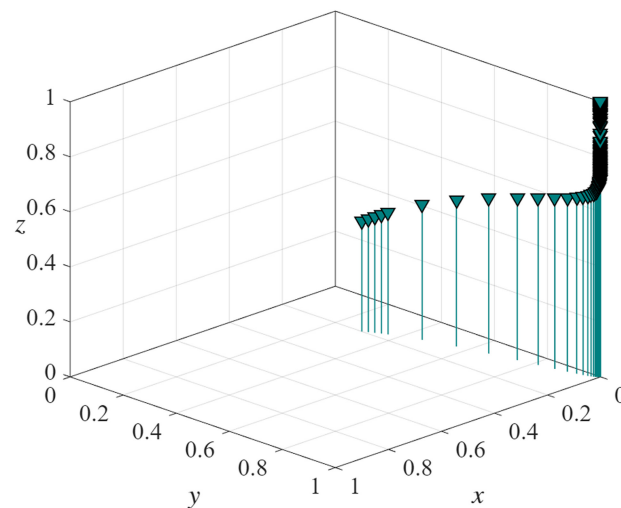


FIGURE 6 | Evolutionary dynamics converging to $E7(0, 1, 1)$.



approaches the edge (0, 1, 1) of the unit cube. Under this equilibrium, treatment plants bear the full burden of PPCPs removal from heavily contaminated influent, which elevates operational costs and may reduce removal efficiency compared to scenarios with concurrent source control.

5.3 | Sensitivity Analysis

To explore how variations in key parameters influence system dynamics and equilibrium selection, this section presents sensitivity analysis focusing on the voluntary cooperation scenario **E5(1,1,0)**, which represents an attractive policy target given its achievement of environmental objectives through market mechanisms rather than regulatory coercion. Two parameters with substantial practical and policy significance receive detailed examination: the clean production cost C_1 and the clean production revenue R_1 .

Figure 7 illustrates how different values of C_1 affect evolutionary trajectories in phase space, with C_1 varying across 0.3, 0.4 and 0.5 (scaled by factor 100). When clean production costs remain low ($C_1 = 30$), the profitability advantage $(R_1 - C_1) - (R_2 - C_2)$ increases, accelerating enterprise convergence toward clean production and hastening system arrival at E 5. As C_1 rises to 40 and subsequently to 50, the net benefit differential narrows, slowing the rate of enterprise strategy evolution and potentially threatening equilibrium stability if costs rise sufficiently to reverse the profitability ranking. The three trajectories in Figure 7 demonstrate progressively slower convergence as C_1 increases, with the highest cost scenario exhibiting the most gradual approach to the stable equilibrium. This sensitivity pattern carries important policy implications: government efforts to reduce clean production costs through technology transfer programs, green manufacturing subsidies, or research and development support can accelerate voluntary cooperation emergence and strengthen equilibrium stability against parameter fluctuations.

Figure 8 examines the impact of varying clean production revenue R_1 across values of 1.0, 1.2, and 1.4 (scaled by factor 100). Higher values of R_1 correspond to stronger green brand

premiums, more pronounced consumer preferences for environmentally responsible products, or enhanced market access for certified sustainable manufacturers. The simulation results reveal that elevated R_1 values substantially accelerate enterprise convergence toward clean production, as the widening profitability gap $(R_1 - C_1)(R_2 - C_2)$ creates stronger evolutionary pressure favoring cooperative strategy adoption. When $R_1 = 100$ (the lowest value examined), enterprise clean production remains profitable but the convergence trajectory exhibits relatively gradual dynamics. Increasing R_1 to 120 and 140 progressively steepens the trajectory slope, indicating faster strategy adjustment rates and more robust equilibrium stability. These findings suggest that policies enhancing green product market value, such as eco-labeling programs, green public procurement preferences, or environmental certification standards, can effectively promote voluntary private sector cooperation in PPCPs governance without requiring costly regulatory enforcement infrastructure.

5.4 | Sensitivity Analysis of Treatment Plant Parameters

The sensitivity analysis of treatment plant parameters employs equilibrium $E_7(0,1,1)$ as the convergence target, representing a governance configuration wherein pharmaceutical enterprises persist with traditional production methods while wastewater treatment plants universally adopt upgraded technologies under sustained government regulatory oversight. This equilibrium selection reflects realistic scenarios where enterprise clean production transition faces substantial economic barriers that render voluntary source reduction unattainable within policy-relevant time horizons, yet environmental protection objectives can nonetheless be achieved through intensified end-of-pipe treatment coupled with persistent regulatory engagement. From a stability analysis perspective, E_7 achieves asymptotic stability when the combined effect of treatment plant subsidies, penalties, and reputational losses exceeds the intrinsic cost disadvantage of technology upgrading, while simultaneously the enterprise clean production incentive remains insufficient to induce behavioral transition away from traditional production. The theoretical significance of E_7 lies in its characterization of

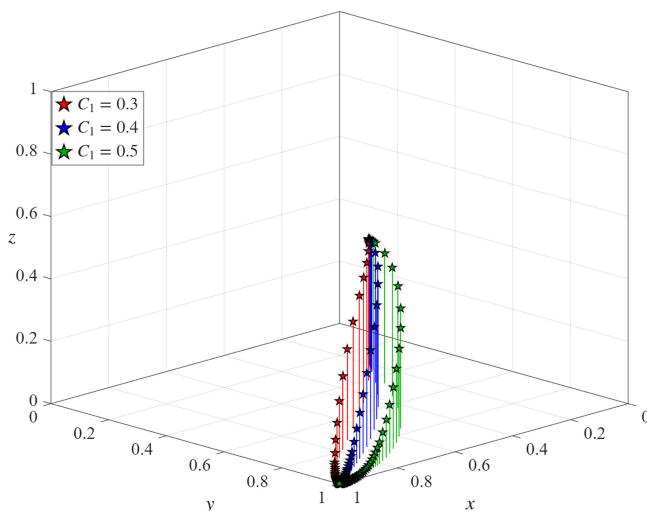


FIGURE 7 | Sensitivity analysis of clean production cost C_1 .

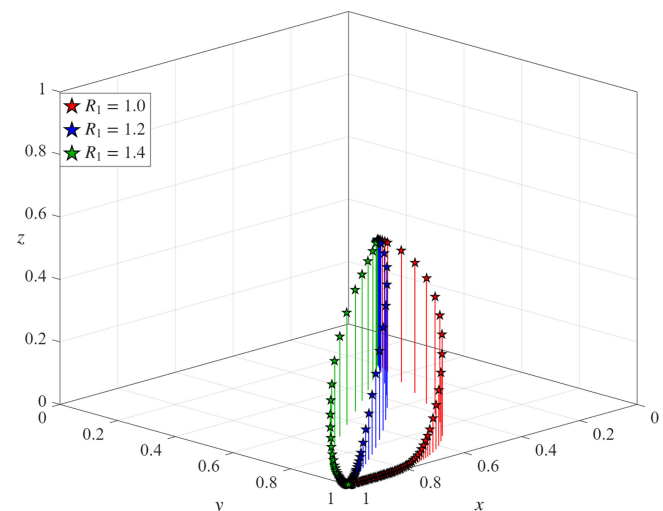


FIGURE 8 | Sensitivity analysis of clean production revenue R_1 .

governance systems where downstream remediation compensates for upstream pollution generation, constituting a second-best outcome that accepts continued enterprise emissions while ensuring adequate treatment capacity to protect receiving water quality. This equilibrium configuration carries particular policy relevance for jurisdictions where pharmaceutical industry competitiveness concerns constrain regulatory stringency on production processes, necessitating alternative governance pathways that achieve environmental objectives through treatment infrastructure investment rather than source control mandates. Examining parameter sensitivity around E_7 therefore reveals the instrument magnitudes required to ensure treatment plant cooperation even when upstream source control fails, providing critical guidance for designing resilient governance frameworks that achieve environmental protection despite incomplete stakeholder coordination.

Figure 9 depicts the evolutionary trajectories under three treatment plant subsidy scenarios converging toward equilibrium $E_7(0,1,1)$: $S_2=10$, $S_2=25$, and $S_2=40$, representing low, moderate, and high fiscal support intensities respectively. The three-dimensional phase space visualization tracks the joint evolution of enterprise clean production probability, treatment plant technology upgrading probability, and government active regulation probability from identical initial conditions across subsidy variations. As illustrated, all three trajectories converge toward the region characterized by low enterprise cooperation, high treatment plant upgrading, and sustained government regulation, confirming the asymptotic stability of equilibrium E_7 across the examined subsidy range. The enterprise clean production probability declines toward zero because the parameterization reflects scenarios where traditional production remains more profitable than clean alternatives even under regulatory pressure, causing the enterprise population to evolutionarily drift toward non-cooperative strategies regardless of treatment plant subsidy variations. However, the treatment plant dimension exhibits markedly different dynamics across subsidy treatments: the low subsidy scenario ($S_2=10$, red markers) produces a trajectory with more dispersed vertical distribution in the phase space, indicating that treatment plant upgrading probability requires extended evolutionary periods to approach unity when

fiscal incentives are modest. The high subsidy scenario ($S_2=40$, green markers) generates a more concentrated trajectory cluster with substantially higher point density, demonstrating that generous fiscal support accelerates treatment plant behavioral convergence even when upstream enterprises fail to cooperate. The moderate subsidy scenario ($S_2=25$, blue markers) displays intermediate characteristics, with trajectory density and convergence speed falling between the low and high subsidy cases, suggesting a monotonic relationship between subsidy intensity and convergence acceleration within this parameter range. The government regulation probability maintains elevated values throughout the evolutionary process across all subsidy treatments, reflecting the theoretical prediction that regulatory persistence becomes necessary when enterprise cooperation cannot be achieved, as the penalty threats and subsidy disbursements inducing treatment plant cooperation require continued government engagement to remain credible and effective.

Figure 10 examines the sensitivity of evolutionary dynamics to the penalty intensity F_2 imposed on treatment plants maintaining conventional technologies under active regulatory oversight, with trajectories again converging toward equilibrium $E_7(0,1,1)$. Three penalty magnitudes are analyzed: $F_2=5$, $F_2=20$, and $F_2=35$, spanning the range from nominal enforcement consequences to stringent regulatory sanctions. The trajectory patterns reveal that penalty intensity exerts substantial influence on treatment plant behavioral dynamics, complementing the subsidy effects documented in Figure 9 through a distinct incentive mechanism that increases the expected cost of non-compliance rather than reducing the cost of cooperation. The low penalty scenario ($F_2=5$, red markers) generates a trajectory characterized by gradual ascent in the treatment plant upgrading dimension and wider spatial dispersion, indicating that weak enforcement consequences provide insufficient deterrent pressure to accelerate technology adoption beyond baseline adjustment rates. Treatment plants facing minimal penalty exposure rationally delay costly technology investments, producing slower convergence dynamics even when government maintains active regulatory presence. The moderate penalty scenario ($F_2=20$, blue markers) demonstrates noticeably faster convergence with trajectory points

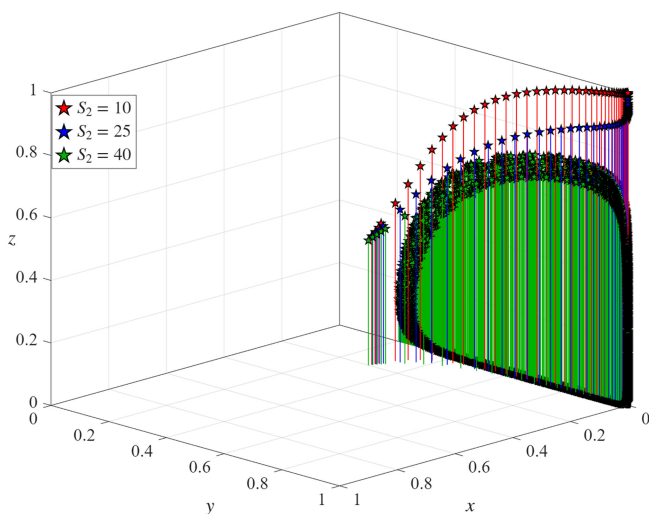


FIGURE 9 | Sensitivity analysis of treatment plant subsidy S_2 .

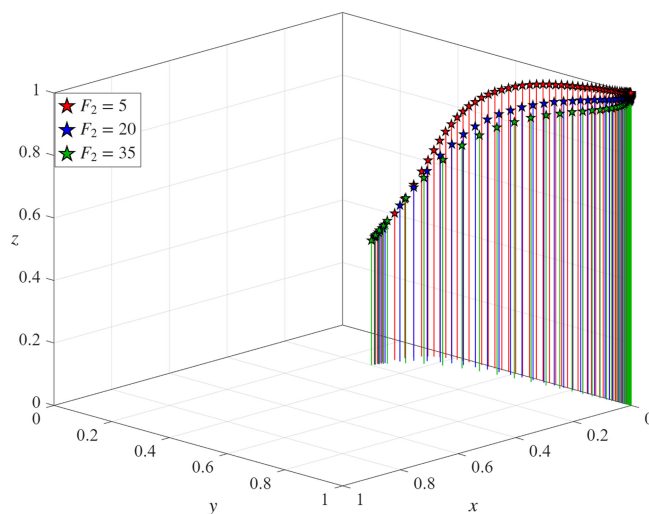


FIGURE 10 | Sensitivity analysis of treatment plant penalty F_2 .

concentrated at higher upgrading probability values, confirming that meaningful enforcement consequences can effectively modify treatment plant behavioral dynamics when calibrated at sufficient magnitudes. The high penalty scenario ($F_2 = 35$, green markers) produces the most rapid and concentrated convergence trajectory, with treatment plant upgrading probability reaching elevated values within fewer evolutionary periods, as the substantial penalty exposure renders conventional technology maintenance economically untenable under regulatory scrutiny. The comparative analysis across penalty treatments reveals behavioral mechanisms that complement subsidy effects documented in Figure 9: whereas subsidies reduce the absolute cost of technology upgrading by providing fiscal support for capital investment, penalties increase the expected cost of maintaining conventional treatment operations through regulatory sanctions, and both instruments shift the payoff differential in favor of technology upgrading adoption. The combined evidence from Figures 9 and 10 supports policy recommendations emphasizing coordinated deployment of both positive and negative incentive instruments, with subsidies potentially more effective during initial adoption phases when capital barriers constitute the primary obstacle, while penalties become increasingly potent for inducing behavioral change among recalcitrant operators who have not responded to fiscal incentives alone.

6 | Discussion

The tripartite evolutionary game model developed in this study reveals the strategic interaction mechanisms among pharmaceutical enterprises, wastewater treatment plants, and government environmental protection agencies in PPCPs pollution control, and this theoretical contribution engages productively with empirical findings from the emerging contaminant governance literature. Model analysis demonstrates that government subsidies and penalty intensities exert decisive influence over wastewater treatment plant technology upgrading decisions, rather than upstream pharmaceutical enterprise clean production behavior alone determining such choices. Empirical evidence lends strong support to this conclusion: membrane bioreactor and other advanced treatment processes achieve markedly different antibiotic removal efficiencies across seasons, with biodegradation efficiency reaching 64.79% under elevated summer temperatures while winter microfiltration retention of antibiotics remains limited, and these observations indicate that wastewater treatment plants must weigh operational costs against actual removal performance when selecting treatment technologies (Zheng et al. 2019). Ecological risk assessments of micropollutants in wastewater treatment plant effluents have simultaneously revealed that perfluorooctanoic acid, bisphenol A, and other emerging contaminants remain excluded from current effluent regulatory frameworks, a finding that underscores the necessity of proactive government regulatory strategies (Liu et al. 2024). Monitoring data from Himalayan foothill wastewater treatment plants deserve particular attention because they document negative removal phenomena for certain PPCPs, wherein effluent concentrations exceed influent levels, and this pattern proves especially pronounced for ciprofloxacin, carbamazepine, and caffeine, thereby corroborating this model's core assumption that conventional treatment processes cannot

effectively address PPCPs contamination (Silori et al. 2025). Life cycle assessment studies of advanced treatment technologies have further demonstrated that ozone oxidation and granular activated carbon adsorption can substantially improve PPCPs removal rates, yet the indirect environmental impacts arising from energy consumption and resource demands may exceed direct water quality improvement benefits, and this evidence provides a scientific basis for governments to formulate differentiated subsidy policies (Risch et al. 2022).

Stability analysis of the model reveals another important finding: the complete cooperation equilibrium $E_8(1,1,1)$ cannot theoretically achieve asymptotic stability, which implies that even after private sector actors have fully adopted environmentally friendly strategies, governments still face fiscal incentives to withdraw from regulatory engagement. This theoretical finding warrants contextualization within broader governance realities. The model specification captures fiscal incentives that induce regulatory withdrawal when compliance objectives have been achieved, a phenomenon documented in environmental policy implementation where agencies reallocate enforcement resources toward emerging priorities once established programs demonstrate success. However, real-world regulatory systems operate under additional constraints that the stylized model does not fully incorporate. Legal mandates embedded in environmental legislation create binding obligations for continued monitoring and standard enforcement regardless of current compliance rates, insulating regulatory agencies from purely fiscal optimization pressures. Bureaucratic reputation concerns motivate sustained regulatory presence because agency credibility and future budget allocations depend on demonstrated ongoing engagement rather than historical accomplishments. Political accountability mechanisms, including legislative oversight and public expectations shaped by environmental advocacy organizations, constrain government discretion to reduce regulatory intensity even when immediate fiscal calculations might favor withdrawal. These institutional factors suggest that E_8 instability represents a theoretical limiting case that identifies governance vulnerabilities requiring institutional safeguards, rather than an inevitable empirical prediction. Policy designs incorporating statutory monitoring requirements, earmarked enforcement funding, and transparent compliance reporting can mitigate the regulatory withdrawal incentives that the model identifies, thereby approximating stable complete cooperation outcomes in practice even if pure asymptotic stability remains theoretically unattainable. This theoretical prediction aligns closely with stakeholder survey research conclusions: experts drawn from government, industry, and research institutions generally identify advanced wastewater treatment technologies, healthcare professional education to reduce overprescription, pharmaceutical return programs coupled with public education, and mandatory secondary treatment capacity requirements for all municipal authorities as the most effective management strategies, and the successful implementation of these strategies depends on sustained government regulatory commitment rather than one-time policy interventions (Doerr-MacEwen and Haight 2006). Regarding public participation dimensions, survey research among residents in Xiamen, China has found significant willingness to pay for PPCPs pollution control policies, with income level, environmental concern, and subjective financial health status serving as important predictors of payment

willingness, and these findings suggest that governments can draw upon a foundation of social support when implementing active regulatory strategies (Wang et al. 2015). Cross-national comparative research on pharmaceutical regulation in reclaimed water has noted that despite ongoing monitoring efforts in multiple developed countries, pharmaceutical contaminants have generally not been incorporated into water quality regulatory systems, and Israel, the country with the highest reclaimed water utilization rate globally, similarly lacks systematic regulation of pharmaceutical residues in its wastewater, a situation that effectively constitutes an epidemiological experiment conducted on citizens and the environment (Miarov et al. 2020). Urban wastewater monitoring data from China during the COVID-19 pandemic further confirm this study's theoretical inference regarding the impacts of public health emergencies on PPCPs discharge: following policy adjustments, ibuprofen and paracetamol concentrations surged to 67-fold and 311-fold pre-pandemic levels respectively; direct discharge of high pharmaceutical-load wastewater can pose significant ecological risks to aquatic organisms, and establishing comprehensive wastewater treatment systems proves particularly critical for remote and underdeveloped regions (Yu et al. 2025).

The policy instruments identified as effective levers for promoting cooperative behaviors nonetheless carry implementation risks and may trigger unintended stakeholder conflicts that warrant explicit consideration. Differentiated subsidy programs targeting treatment plant technology upgrading face administrative challenges in establishing credible eligibility criteria, monitoring actual technology deployment, and preventing strategic gaming wherein facilities claim subsidies without achieving intended pollution reduction outcomes. Fiscal sustainability concerns arise when subsidy magnitudes sufficient to induce widespread technology adoption impose significant burdens on government budgets, potentially competing with other environmental protection priorities or triggering taxpayer resistance that undermines political support for continued program funding. Penalty enforcement mechanisms require robust monitoring capacity and political willingness to impose consequences on non-compliant actors, conditions that may prove difficult to sustain when regulated enterprises possess significant economic or political influence. The pharmaceutical industry in particular may mobilize lobbying resources to resist stringent clean production requirements, arguing that elevated compliance costs threaten domestic production competitiveness and employment, thereby creating political economy dynamics that constrain regulatory ambition. Treatment plants facing technology upgrading mandates without adequate fiscal support may experience financial distress that compromises service quality for conventional pollutant removal, potentially creating unintended trade-offs between PPCPs control and traditional wastewater treatment performance. Small and medium-sized enterprises with limited capital access may disproportionately bear clean production transition burdens, raising equity concerns that complicate policy design and implementation. These challenges suggest that effective PPCPs governance requires not only well-calibrated economic instruments but also attention to stakeholder consultation processes, phased implementation schedules, and complementary support mechanisms that facilitate behavioral transitions while managing distributional consequences.

7 | Conclusion

This study constructs a tripartite evolutionary game model encompassing pharmaceutical enterprises, wastewater treatment plants, and government environmental protection agencies, thereby systematically revealing the internal mechanisms and evolutionary patterns of multi-stakeholder strategic interactions in PPCPs pollution control. Research findings demonstrate that when clean production and technology upgrading possess inherent profitability advantages, market mechanisms can drive private sector actors to spontaneously converge toward the environmentally friendly equilibrium $E5(1,1,0)$ in the absence of government regulatory intervention, and under such circumstances governments may appropriately withdraw from direct regulation to conserve administrative resources. The complete cooperation equilibrium $E8(1,1,1)$, however, cannot achieve asymptotic stability due to the presence of a positive eigenvalue, which means that once private sector actors have fully adopted cooperative strategies, governments will face fiscal incentives to reduce regulatory investment, thereby threatening the long-term maintenance of cooperative configurations. The model further reveals that government subsidies and penalty intensities primarily govern wastewater treatment plant technology upgrading decisions, rather than upstream enterprise clean production behavior directly determining such choices, and this finding provides a theoretical basis for precisely targeted policy interventions. The mathematical finding that upstream enterprise production choices do not directly alter treatment plant upgrading incentives through cost transmission channels reflects the substantive economic reality that both technology options face equivalent pollution burden increases, revealing that market-based coordination cannot spontaneously emerge without regulatory intervention. Sensitivity analysis confirms that reductions in clean production costs and increases in green product revenues can significantly accelerate system convergence toward ideal equilibria, indicating that technological progress and market incentives serve as critical leverage points in PPCPs governance. Additional sensitivity analysis of treatment plant subsidy and penalty parameters demonstrates that both positive fiscal incentives and negative regulatory sanctions effectively promote technology adoption, with policy designers possessing flexibility to select instrument combinations based on fiscal constraints and political feasibility considerations.

Building upon these theoretical findings, this study advances the following policy recommendations. Governments should first establish differentiated fiscal incentive systems that provide targeted subsidies to wastewater treatment plants adopting advanced treatment technologies such as ozone oxidation, granular activated carbon adsorption, or membrane separation, while simultaneously linking subsidy magnitudes to actual PPCPs removal efficiencies, because this approach can overcome the inhibitory effects of high capital investment requirements on technology upgrading. For pharmaceutical enterprises, this study recommends enhancing the economic returns from clean production through market-based instruments including green certification labels and preferential government procurement of environmentally friendly products, rather than relying solely on end-of-pipe penalties, since this approach can stimulate intrinsic enterprise emission reduction motivation while reducing regulatory costs. Given the structural instability of complete

cooperation equilibrium, governments must maintain necessary monitoring capacity and enforcement deterrence even after widespread private sector compliance has been achieved, and adopting a low-cost regulatory model that combines random inspection with information disclosure can prevent system regression toward non-cooperative states triggered by regulatory relaxation. Finally, considering that public health emergencies may cause dramatic fluctuations in specific pharmaceutical discharges, this study recommends reserving emergency treatment redundancy in wastewater treatment facility planning and incorporating wastewater pharmaceutical monitoring into routine public health surveillance systems, as these measures can enhance the resilience and adaptive capacity of water environment governance systems.

Several limitations of this study merit acknowledgment and suggest directions for future research. The binary strategy specification, while enabling tractable mathematical analysis, simplifies the continuous spectrum of clean production intensity and technology upgrading options available to real-world decision-makers; future extensions could incorporate multi-level strategy spaces or continuous investment variables that better capture the incremental nature of environmental improvement decisions. The tripartite model structure excludes potentially important actors including pharmaceutical consumers whose purchasing behaviors shape market incentives for green products, health-care institutions whose prescribing practices influence total pharmaceutical consumption and environmental loading, and civil society organizations whose advocacy activities affect regulatory stringency and public awareness. Parameter calibration relies on stylized approximations drawn from diverse empirical sources rather than systematic data collection targeting the specific PPCPs governance context, limiting the precision of quantitative predictions regarding convergence speeds and equilibrium selection probabilities. The static payoff specification assumes fixed cost and benefit parameters throughout the evolutionary process, whereas real-world technological learning curves, market development trajectories, and regulatory capacity building would introduce parameter dynamics that could substantially alter long-run system behavior. The model treats each stakeholder group as homogeneous populations, neglecting heterogeneity in firm size, technology vintage, regulatory jurisdiction, and other factors that generate differential responses to policy instruments within each category. These limitations notwithstanding, the analytical framework developed herein provides valuable qualitative insights into the strategic interdependencies characterizing PPCPs governance systems and offers a foundation for more detailed empirical and computational investigations.

Author Contributions

Xue Lei: conceptualization, writing – original draft, formal analysis, methodology. **Yuying Luo:** writing – review and editing, methodology. **Fengmei Liu:** formal analysis, writing – review and editing. **Xueguo Xu:** conceptualization, writing – review and editing, project administration, supervision.

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The authors have nothing to report.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. The numerical simulation results, including all figures presented in this manuscript, were generated using MATLAB R2021a based on the theoretical model parameters specified in Tables 1 and 2. The simulation code and parameter configurations used to generate Figures 1–10 can be made available upon request to facilitate reproduction of results.

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